Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

Arctic – Monitoring Forecast Centre
ARCTIC_MULTIYEAR_PHY_ICE_002_016

Issue: 1.0

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CHANGE RECORD

When the quality of the products changes, the QuID is updated and a row is added to this table. The third column specifies which sections or sub-sections have been updated. The first column should mention the version of the product to which the change applies.

Issue	Date	§	Description of Change	Author	Validated By
1.0	August 2024	All	Creation of the document	T. Williams	L. Bertino

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document covers the product **ARCTIC_MULTIYEAR_PHY_ICE_002_016**, which is one of the two regional multiyear products of physical sea ice parameters in the Arctic. It covers the period from 1991 to the year before present, is extended regularly every 6 months, and is produced by the neXtSIM-F system. This system is based on neXtSIM (neXt-generation Sea Ice Model), with the "F" standing for "forecast" and encapsulated pre- and post-processing done to perform assimilation and to create netcdf files to be distributed through CMEMS (for example). An earlier version of the system was presented by Williams et al (2021). The 2-dimensional sea ice variables delivered to CMEMS are: sea-ice concentrations (total, young ice, and multi-year), thicknesses (total, young ice, and multi-year), drift speed, snow depths, albedo, age and ridged ice volume fraction. The exact nomenclature used in the files is given in the associated product user manual (PUM).

Two datasets are available for this product through the CMEMS Catalogue:

- daily-averaged fields: *cmems_mod_arc_phy_my_nextsim_P1D-m* and;
- monthly-averaged fields: *cmems_mod_arc_phy_my_nextsim_P1M-m*.

More technical details are available in the associated PUM. Both datasets are provided on a North polar stereographic grid at 3 km resolution, covering the Arctic area of responsibility North of 63N, with some margin out to 53 N. The quality information statistics are computed on the area of responsibility North of 63N. The present QuID presents summary statistics (Estimated Accuracy Numbers, EANs) and maps.

I.2 Summary of the results

Details of the validation procedure are given in section V ("Validation framework"), while full results are given in section VI ("Validation results"). Table 1 also gives average errors for the full validation period (depends on availability of validation data). These results are summarized as follows:

- Sea ice concentration (assimilated)
  - Mean modeled concentration is slightly higher (by about 4-8%) than the passive microwave estimates. This translates to a slightly higher sea ice area.
  - Modelled extent is slightly under (by about 10^5 km^2) the extent estimated from passive microwave.
  - The extent bias and integrated ice edge error (IIEE) is greatest from October to December (about -0.5 and 1Mkm^2 respectively).
- Sea ice thickness (assimilated from October 2010)
  - When compared to CS2-SMOS, the errors are greatest in October (about 0.4-0.6m), dropping to the level of the observation errors (about 0.2-0.3m) after about 4-5 weeks.
  - When compared to the PIOMAS reanalysis, the neXtSIM mean thickness is somewhat lower, especially in the 1990s (it is intially lower by 0.4m, but the difference is only 0.2m in the late 2000s). The mean difference between the two models is quite low (about 0.1m) after the neXtSIM product starts assimilating CS2-SMOS in 2010.

- Sea ice age
  - The mean age is lower by about a quarter of a year and the root mean square difference (RMSD) of 0.8 years reflects a slightly different spatial distribution although this is probably acceptable given the different methods of estimating the age in the model and the passive-microwave-derived product.
- Surface albedo
  - On average the model is overestimating the albedo by 0.08 while the RMSD is 0.15.
- Multi-year extent
  - The model is overestimating the extent by a moderate amount (0.5-1 Mkm^2), and the high IITE (integrated ice type error) of 1.45 Mkm^2 shows there is some displacement error between the modelled and observed areas of multi-year ice.
- Sea ice drift
  - The drift bias is very low (80-90m) compared to both the 24-h and 48-h OSI-SAF products
  - The trajectory error compared to the 24-h OSI-SAF is good on average (4.19 km/day) although it is higher in the winter when the observations are more reliable. In the summer, when the observation product needs to be extrapolated using a simple drift model ("free drift" - assuming the ice has no internal stresses that can resist winds and currents) the errors are lower (probably because this model and our one use the ERA5 reanalysis as wind input). Our drift is very good compared to the 48-h OSI-SAF product (3.2 km/day) (only compared in winter).

I.3 Estimated Accuracy Numbers

See section V for explanations of the validation products and the error metrics, and section VI for the full results.

***Table 1**: Estimated accuracy numbers. Note two sources of validation are used for "Sea Ice Thickness"*

Variable	Region	Supporting observations	RMSD	Bias (model-observation)
Sea ice concentration	Full domain	SSMIS	0.15	0.04
Sea ice thickness	Full domain	CS2-SMOS	0.28 m	0.02 m
Sea ice thickness	Full domain	PIOMAS reanalysis	0.51 m	-0.19 m
Sea ice age	Full domain	NERSC	0.81 years	-0.25 years
Surface albedo	Full domain	OLCI	0.15	0.08
			IIEE	**Bias (model-observation)**
Sea ice extent	Full domain	SSMIS	0.37 M km^2	-0.08 M km^2
			IITE	**Bias (model-observation)**
Multi-year ice extent	Full domain	SSMIS	1.45 M km^2	0.44 M km^2
			Trajectory error	**Bias (model-observation)**
Sea ice drift displacement (24h)	Full domain	Passive microwave	4.19 km	-0.08 km
Sea ice drift displacement (48h)	Full domain	Passive microwave	6.37 km	0.09 km

II PRODUCTION SYSTEM DESCRIPTION

Production centre's name

Nansen Environmental and Remote Sensing Center (NERSC), Bergen, Norway.

Production system's name

Arctic sea ice multiyear product
(CMEMS name: ARCTIC_MULTIYEAR_PHY_ICE_002_016)

Production centre description for the version covered by this document

The Arctic Sea Ice Reanalysis system uses the neXtSIM stand-alone sea ice model running the Brittle-Bingham-Maxwell sea ice rheology (Olason et al, 2022) on an adaptive triangular mesh of 10 km average cell length. The model domain covers the whole Arctic domain, from Bering Strait to the North Atlantic. neXtSIM is forced by reanalyzed surface atmosphere forcings (ERA5) from the ECMWF (European Centre for Medium-Range Weather Forecasts) and ocean forcings from TOPAZ4b, the ARC MFC MYP system (002_003). neXtSIM assimilates satellite sea ice concentrations from Passive Microwave satellite sensors, and sea ice thickness from CS2SMOS in winter from October 2010 onwards. The output variables are sea ice concentrations (total, young ice, and multi-year ice), sea ice thickness, sea ice velocity, snow depth on sea ice, sea ice type, sea ice age, sea ice ridge volume fraction and sea ice albedo, provided at daily and monthly frequency. The adaptive Lagrangian mesh is interpolated for convenience on a 3 km resolution regular grid in a Polar Stereographic projection. The projection is identical to other ARC MFC products.

II.1 neXtSIM Production at NERSC

The model results are produced with the neXtSIM stand-alone sea ice model. Daily assimilation is done with a nudging method.

II.2 Observational Data

We assimilate sea ice concentration from OSI SAF (passive microwave using the SSMIS sensor) and CS2-SMOS sea ice thickness when available (in winter from October 2010 onwards).

III VALIDATION FRAMEWORK

In this section we illustrate the model domain, introduce the different error metrics we use in the results section (section VI), describe the different data sources used by the model as inputs, and describe the data we use for validation.

The model domain for which sea ice results are available is displayed in Figure 1. Here, a sample depiction of the sea ice thickness is shown for illustrative purposes.

The general validation procedure for a model variable is to interpolate this variable onto the grid of some corresponding observations, and calculating certain error metrics to quantify the agreement or disagreement.

The metrics we use are

- $Bias = \frac{1}{N}\sum(X_{mod} - X_{obs})$
- The root mean square difference, $RMSD = \sqrt{\frac{1}{N}\sum(X_{mod} - X_{obs})^2}$
- $Extent\ bias = Extent_{model} - Extent_{obs}$
- Integrated ice edge error (IIEE), which is the area where the model predicts ice but the observations say there is no ice, plus the area where the model predicts no ice but the observations say there is ice.
- Integrated ice type error (IITE), which is the area where the model predicts multi-year ice but the observations say there is no multi-year ice, plus the area where the model predicts no multi-year ice but the observations say there is multi-year ice.
- $Drift\ bias = \frac{1}{N}\sum(Drift_{mod} - Drift_{obs})$, where $Drift = \sqrt{(\triangle x)^2 + (\triangle y)^2}$, $\triangle x$ is the x-component of the drift and $\triangle y$ is the y-component of the drift.
- $Trajectory\ error = \sqrt{\frac{1}{N}\sum(\triangle x_{mod} - \triangle x_{obs})^2 + (\triangle y_{mod} - \triangle y_{obs})^2}$

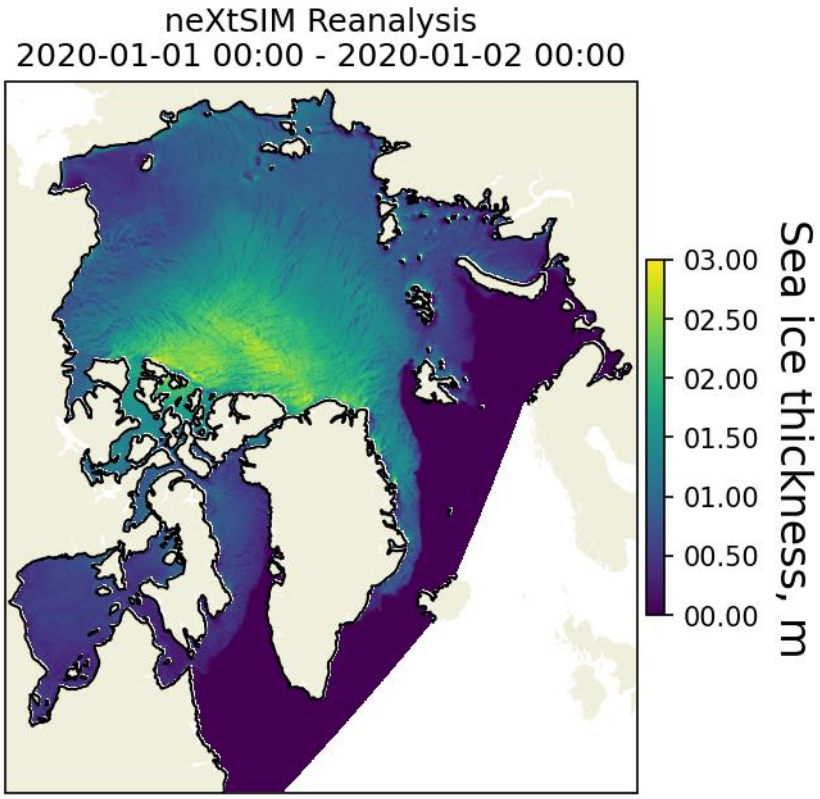

Figure 1: *Map showing an example of sea ice thickness (in m.) from the multi-year ARCTIC_ANALYSISFORECAST_PHY_ICE_002_016 (202411 version). The figure shows the daily mean for 2020-01-01.*

III.1 Model results

The results for the following model variables are validated in this document:

- ✔ Sea ice concentration and extent
- ✔ Sea ice thickness
- ✔ Sea ice age
- ✔ Multi-year ice extent
- ✔ Sea ice albedo
- ✔ Sea ice drift

III.2 External operational products

We use some external products both as direct inputs to the model and as part of the data assimilation procedure during which we modify the model variables at the start of each day. Data used for these purposes are:

- The ERA5 reanalysis is used to provide atmospheric forcing for neXtSIM.
- The TOPAZ4b reanalysis is used to provide oceanic forcing for neXtSIM.

- We assimilate OSI SAF SSMIS sea ice concentration. For 1993-2020 we use version 3 of the climate data record (CDR) from the CDS (climate data store), while for 2021-2023 we use version 3 of the Interim Climate Data Record (ICDR) from the CDS.
- We assimilate CS2-SMOS sea ice thickness (version 2.6 from AWI)
- We use bathymetry from ETOPO (2-arc-minutes resolution) to determine how much (if any) basal stress should be applied to the ice.

III.3 Validation products

The validation products are:

- Sea ice concentration *vs* passive microwave data. This is the OSI-SAF SSMIS concentration which we assimilate (see section III.2). Error metrics:
    1. Concentration bias
    2. Concentration RMSD
- Sea Ice Extent *vs* passive microwave data. This is the OSI-SAF SSMIS concentration which we assimilate (see section III.2) Error metrics:
    1. Bias
    2. IIEE (Integrated ice edge error)
- Sea ice thickness *vs* CS2-SMOS thickness This is the thickness product which we assimilate (see section III.2) It is available from October to April from 2010-10-15 until about 6 weeks before present. Error metrics:
    1. Thickness bias
    2. Thickness RMSD
- Sea ice thickness *vs* PIOMAS reanalysis. Error metrics:
    1. Thickness bias
    2. Thickness RMSD
- Sea ice age *vs* passive-microwave-derived age produced by NERSC (Korosov et al., 2018).
- Age bias
- Age RMSD
- Sea ice type *vs* passive microwave data. This is the OSI-SAF product (version 3 of the CDR from the CDS, available from 1978 to 2020). Error metrics:
    1. MYI extent bias
    2. IITE (integrated ice type error). Like the IIEE it measures the displacement error between modelled and observed regions of MYI.
- Surface albedo vs OLCI-derived surface albedo from U. Bremen. Note the model outputs the sea ice albedo – a surface albedo $a_{surface}$ is produced by a concentration-weighted mean between the sea ice albedo a_{ice} and the ocean albedo (a tuning parameter set to a_{ocean} = 0.07). That is, $a_{surface}=Aa_{ice} + (1-A)a_{ocean}$, where A is the sea ice concentration (area fraction). Error metrics:

    1. Surface albedo bias
    2. Surface albedo RMSD

- Sea ice drift vs passive-microwave-derived 24h product from OSI-SAF (1993-2020). Error metrics:
    1. Bias in drift distance
    2. Trajectory error (distance between the end-points of the modelled and observed trajectories).
- Sea ice drift vs passive-microwave-derived 48h product from OSI-SAF (2018-2021). Error metrics:
    1. Bias in drift distance
- Trajectory error.

Results are provided as tabulated estimated accuracy numbers (summarized in Table 1).

IV VALIDATION RESULTS

IV.1 In this section, we compare the model variables that are provided through CMEMS to available observations. We compare the sea ice concentration and extent to passive microwave data, sea ice thickness to the CS2-SMOS product (blended altimeter and passive microwave measurements), and sea ice age, type and drift to passive-microwave-derived products. Variables that do not have observations to compare to but are included in the product to give a more complete description of the ice state are the ridged ice volume fraction and the young ice concentration.Sea Ice Concentration, Sea Ice Area, Edge Position vs passive microwave data

This data is assimilated daily by the model during this reanalysis.

IV.1.1 Sea ice concentration and area

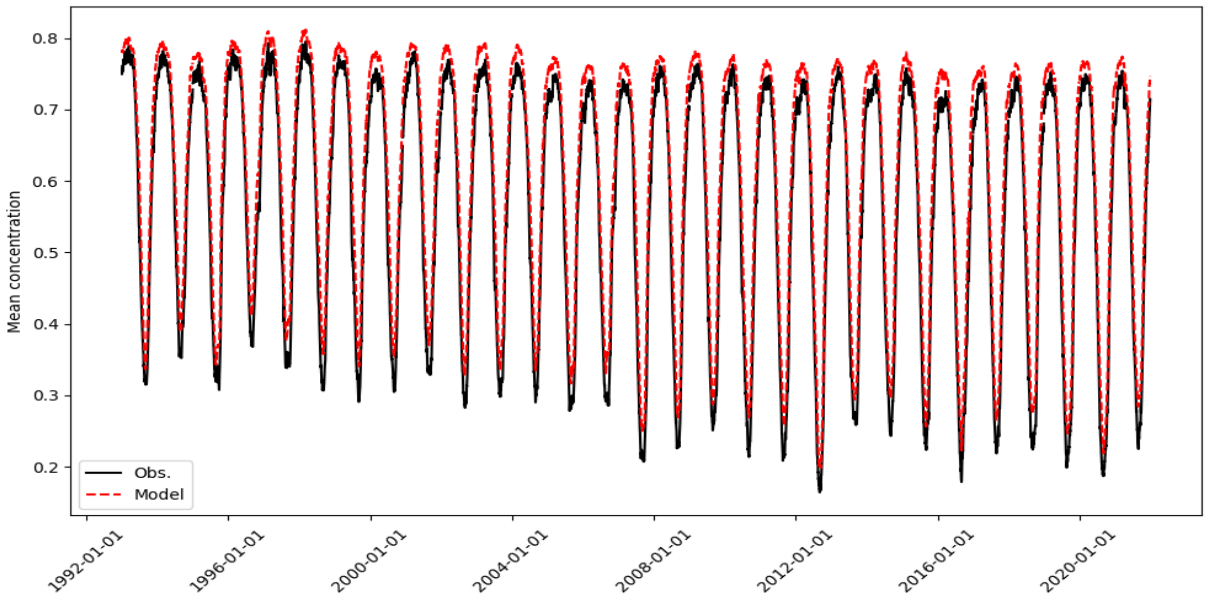

***Figure 2**: Mean sea ice concentration obtained from the model (red dashes) and from satellite observations (back line) (OSI-SAF SSMIS concentration). This is a proxy for sea ice area.*

As seen in Figure 2 (and the concentration bias in Figure 3), the mean model sea ice concentration is slightly higher (by about 4-8%) than the passive microwave estimate, which is known to underestimate the ice concentration. The seasonal cycles match quite well, with the timing of minima and maxima agreeing.

Figure 3 shows that the concentration bias is well within the observation error level. The RMSD is close to the error level (about 10-15%) except in summer when it about 5% above the error level.

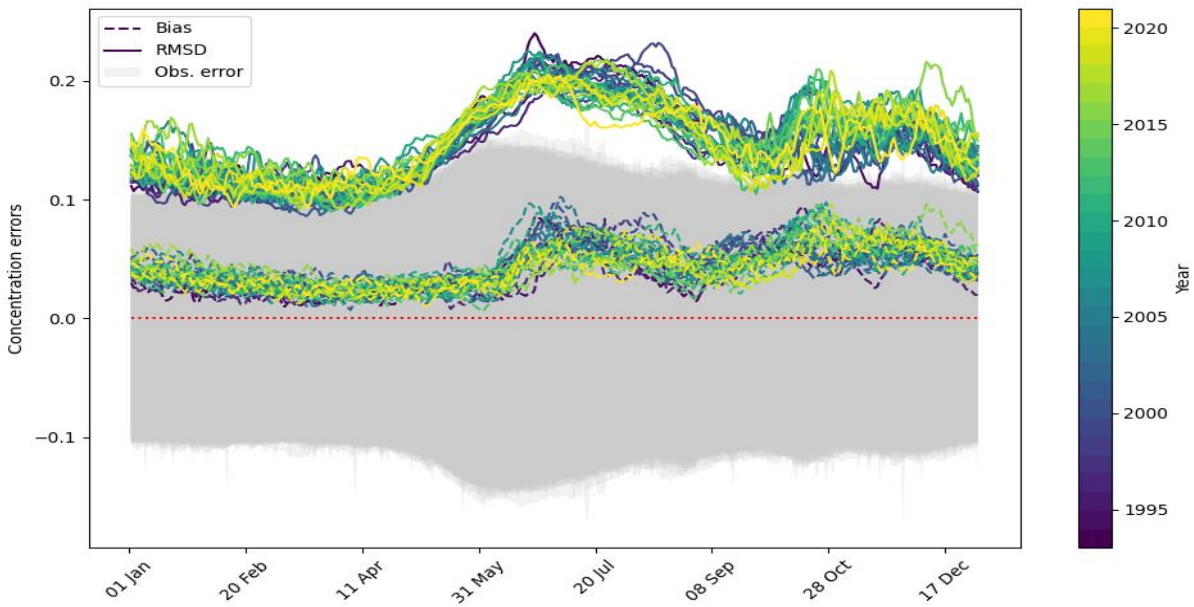

***Figure 3**: Error metrics for sea ice concentration. For reference, a zero bias line is plotted in red.*

IV.1.2 Sea ice extent and ice edge

Figure 4 shows that the bias in extent and IIEE are mostly quite low (the bias is within ±0,1 Mkm^2, while the IIEE is about 0.3-0.6 Mkm^2) except around October to November when the ice-edge is advancing – the model is advancing at a slower speed compared to the observations. Then the bias is about –0.5M km^2 and the IIEE is about 0,5 Mkm^2.

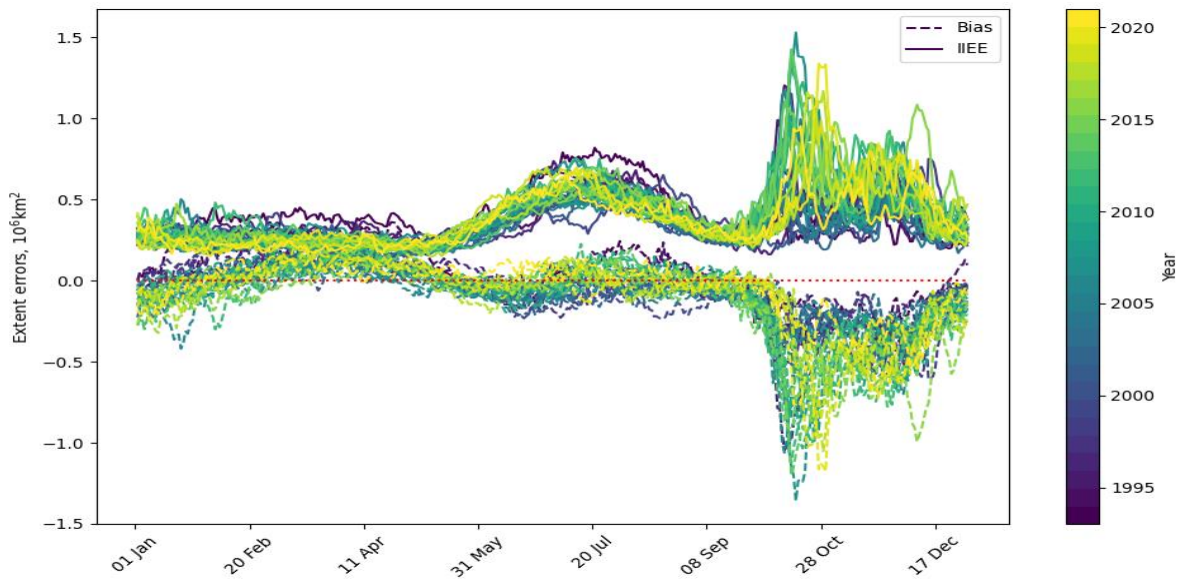

***Figure 4**: Error metrics for sea ice extent. For reference, a zero bias line is plotted in red.*

IV.2 Sea Ice Thickness

Figure 5 shows the error metrics for neXtSIM compared to CS2-SMOS which we assimilate. The bias is quite low (mostly within ±0.2 m) even at the start of winter and notably in 2010 when the thickness data first becomes available. It reduces though after about 4-6 weeks, generally falling within ±0.1 m. The RMSD drops from about 0.4-0.6m at the start of winter to about the error level (about 0.2 m) after this period of 4-6 weeks also.

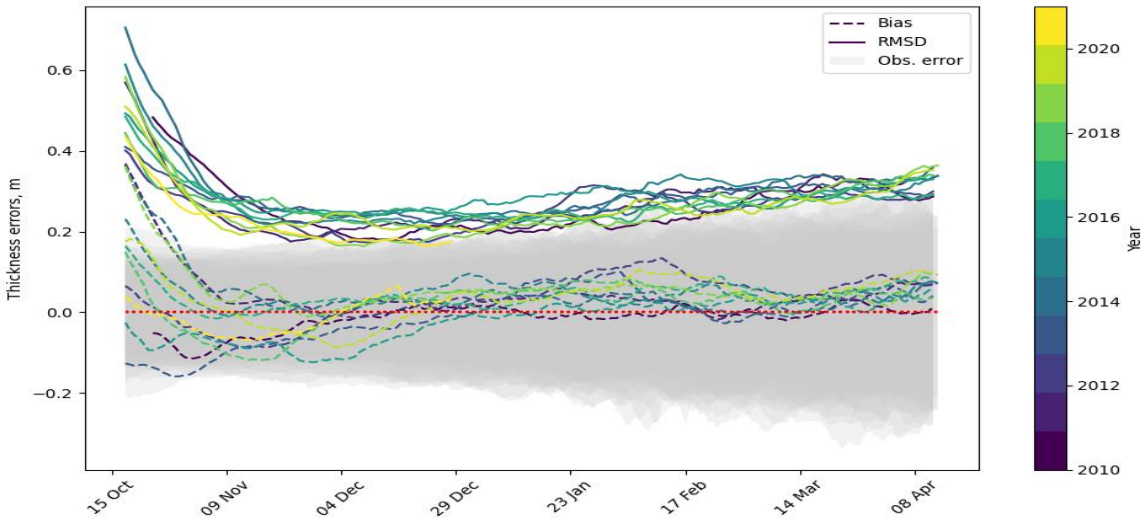

***Figure 5**: Error metrics for sea ice thickness compared to CS2-SMOS. For reference, a zero bias line is plotted in red.*

Figure 6 shows a comparison of the mean thickness between the neXtSIM and the PIOMAS reanalyses. neXtSIM generally gives a lower thickness, especially at the start of the period, when the mean is lower by about 0.4 m. It has a lower rate of decline than PIOMAS so that they are moderately close (the model is about 0.2 m under) by around 2012. They are quite close in the assimilation period, although neXtSIM is still thinner by about 0.1 m.

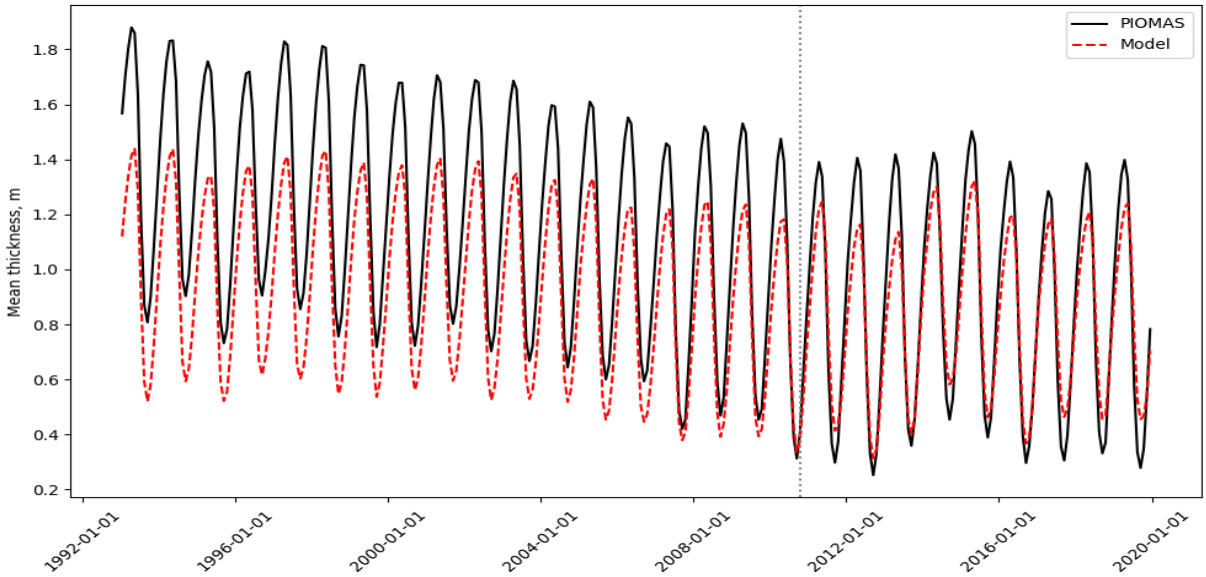

***Figure 6**: Error metrics for sea ice thickness compared to PIOMAS. The vertical gray line marks the time when the CS2-SMOS thickness data starts to be assimilated.*

IV.3 Sea Ice Age

Figure 7 shows that there are relatively large differences between the model and the passive microwave (PM) estimate. Firstly, it is clear that the age is not fully spun-up until about 2000, neither in neXtSIM nor in the PM estimate. After this period the seasonal cycle of the PM estimate has a bigger amplitude than neXtSIM and the timing of the minima and maxima are also different. This could be due to differences in the age calculation – e.g. the PM estimate increases age by 1 year on 15 September while the model increases the age continuously. Also, the model has a volume-weighted average while the PM estimate uses a concentration weighting.

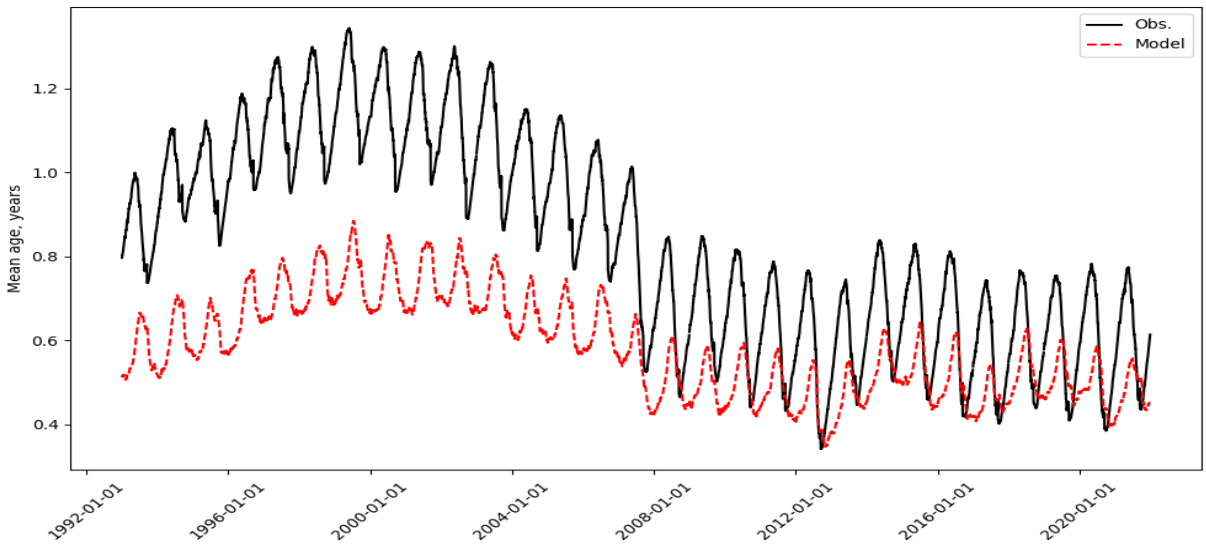

***Figure 7**: Mean sea ice age for neXtSIM and the NERSC passive-microwave-derived age.*

IV.4 Sea Ice Type and Multi-year Ice Extent

Figure 8 shows that the multi-year ice (MYI) extent is higher in the model than in the observation estimate by quite a large amount (0.5 - 1 Mkm^2). This could be mitigated slightly by including the ambiguous ice class in the observed MYI extent (which would increase the estimate from the passive microwave product).

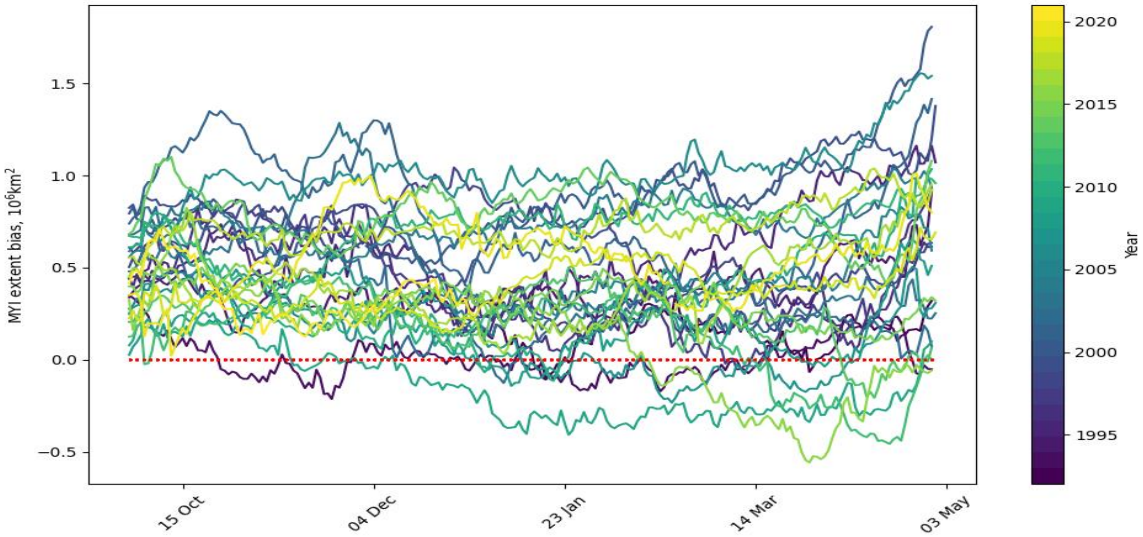

***Figure 8**: MYI extent for neXtSIM and the OSI-SAF passive-microwave-derived ice type. For reference, the dotted red line shows the zero bias line.*

IV.5 Sea Ice Albedo

Figure 9 shows that in May the bias in albedo is within the observation error levels, but in June and July it becomes too high by about 0.1. It drops somewhat in August and September to about 0.05 - 0.06. The RMSD also increases steadily from about 0.1 in May to about 0.2 in June before levelling off and dropping towards the end of the summer, when it is about 0.15. This could be due to melt ponds not being represented in the version of the model that was used.

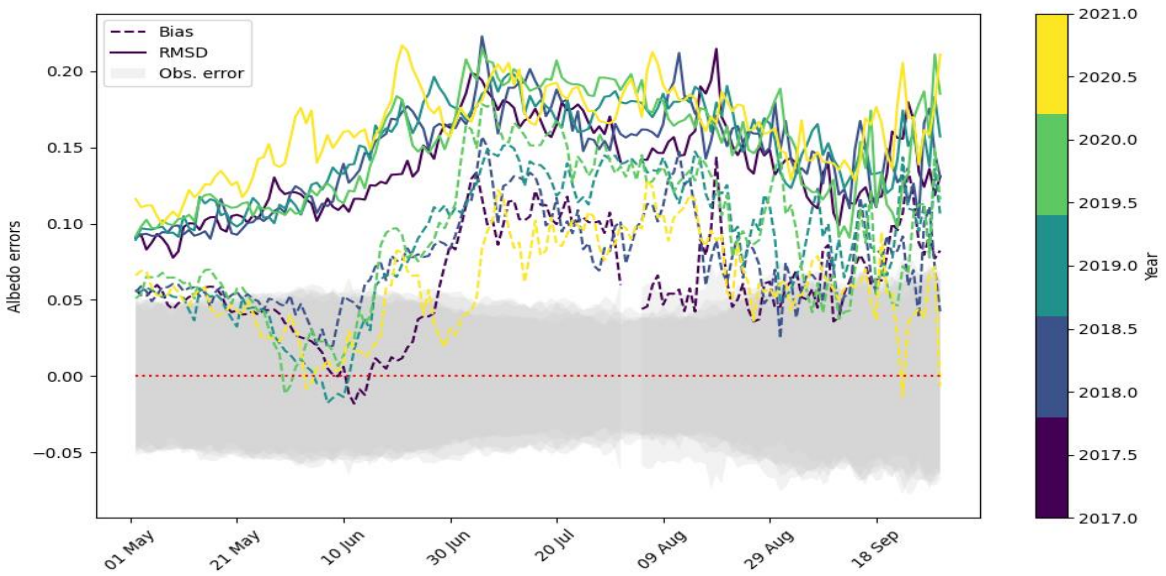

***Figure 9**: Error metrics for the surface albedo. For reference, the dotted red line shows the zero bias line.*

IV.6 Sea Ice Drift

Figure 10shows that the mean drift in the model and the OSI-SAF CDR product is quite close. There is a slight increase in drift speed for both over the period, although the model does not capture the full magnitude of all the events of high drift.

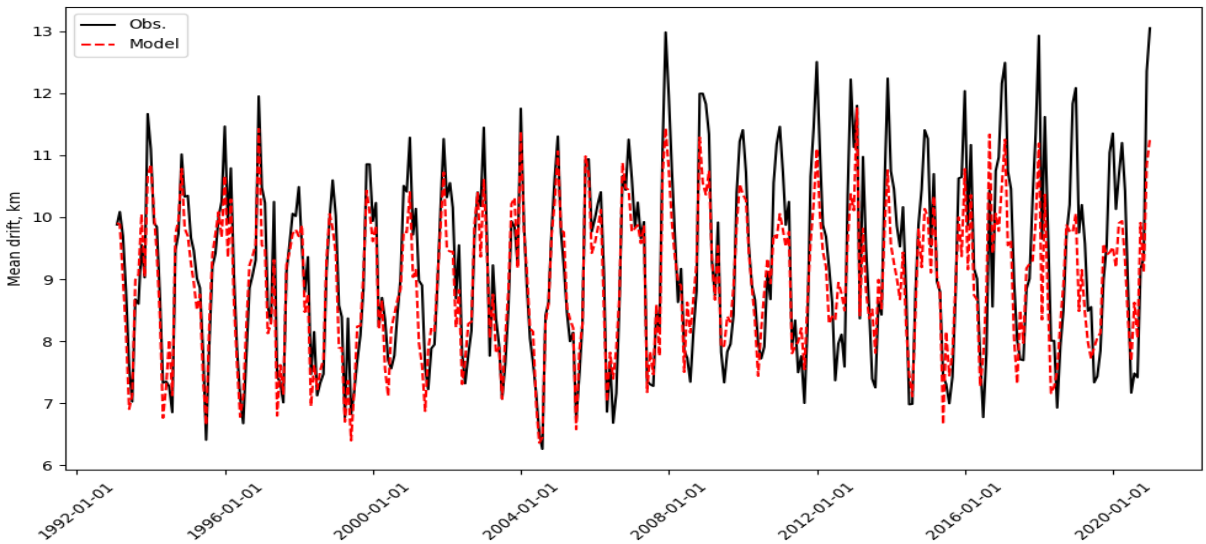

***Figure 10**: RMS sea ice drift for the model and the 24-h OSI-SAF CDR product.*

Figure 11 shows that the bias compared to the OSI-SAF 24-h product is quite low in general (within ±1 km, which is well within the error level of about 3-4 km) while the trajectory error is about 5km in winter, dropping to about 4km in summer. The OSI-SAF product is informed by a

free drift model in summer so that could explain the improved agreement in summer since both are forced by ERA5 winds. The model is also close to free drift in summer since the concentration drops, lowering the ice stiffness.

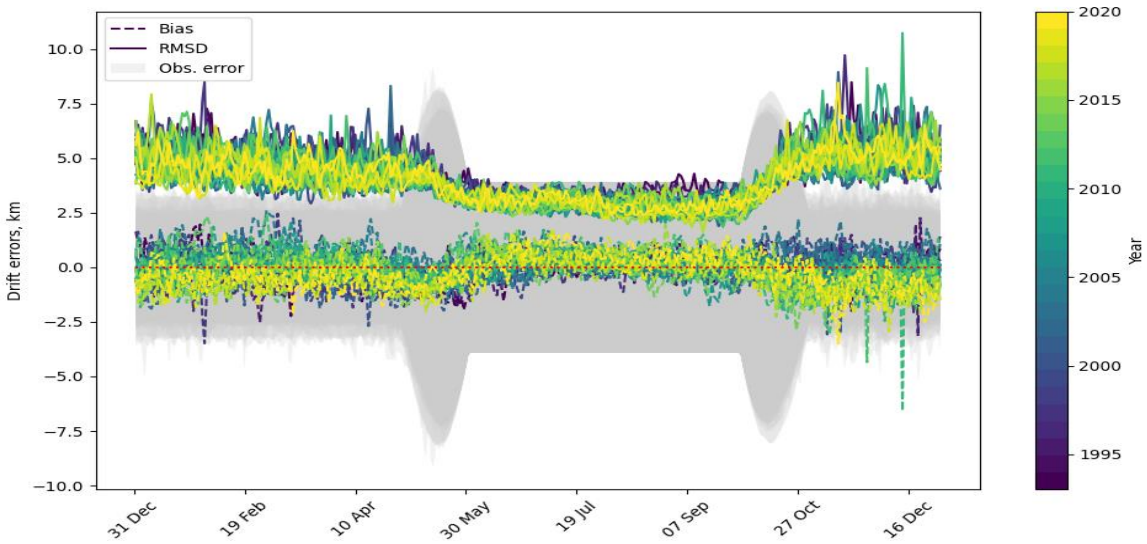

***Figure 11**: Error metrics for the sea ice drift vs the 24-h OSI-SAF CDR product. For reference, a zero bias line is plotted in red.*

Figure 12 shows that the bias compared to the OSI-SAF 48-h product is also quite low (mostly within the error level of ±2km) while the trajectory error is also low, averaging less than 4km/day.

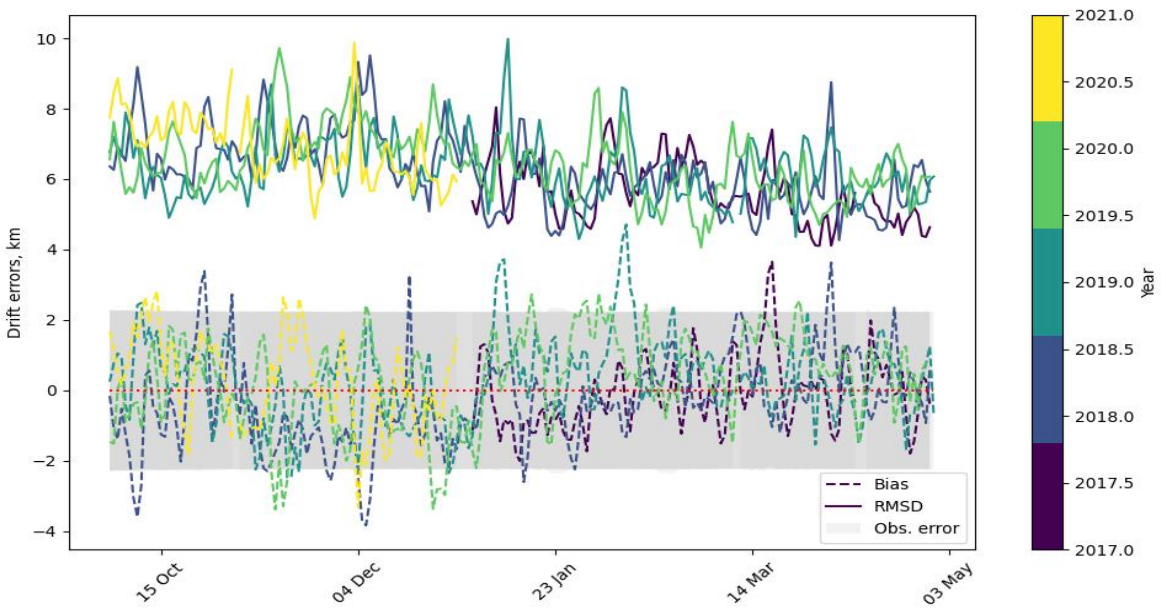

***Figure 12**: Error metrics for the sea ice drift vs the 48-h OSI-SAF product. For reference, a zero bias line is plotted in red.*

IV.7 Conclusion

Overall, the product is performing well, with sea ice drift and thickness closely aligning with observations. The sea ice extent is generally consistent with passive microwave observations, although larger discrepancies occur in October and November when the ice is advancing. Albedo tends to be biased high. Additionally, ice age and multi-year ice extent show reasonable agreement with observational estimates.

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This document describes the first version of the Arctic MFC sea ice reanalysis system.

***Table 7**: List of events, outages and/or changes for each version of the present product.*

Date	Change/Event description	System version	other
27/8/2024	First version of the neXtSIM-F multiyear product, using the Brittle Bingham-Maxwell (BBM) rheology.	1.0	

VI QUALITY CHANGES SINCE PREVIOUS VERSION

August 2024:

Initial version.

VII REFERENCES

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Ólason, E., Boutin, G., Korosov, A., Rampal, P., Williams, T., Kimmritz, M., et al. (2022). A new brittle rheology and numerical framework for large-scale sea-ice models. *Journal of Advances in Modeling Earth Systems*, 14, e2021MS002685.

Williams, T., Korosov, A., Rampal, P., and Ólason, E.: Presentation and evaluation of the Arctic sea ice forecasting system neXtSIM-F, *The Cryosphere*, **15**, 3207–3227, doi:10.5194/tc-15-3207-2021, 2021.